

XAllergen

(eXplainable Allergenicity Prediction of Proteins with web-interface)

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ONE-SENTENCE SUMMARIES

Task: We aimed to build an explainable AI model that predicts protein allergenicity directly from amino acid sequences and delivers results through an interactive web interface.

Approach: We fine-tuned an ESM-2 protein language model and incorporated interpretability techniques, including integrated gradients and attention maps, to highlight key residues, and integrated these functionalities into a user-friendly web application built with Streamlit.

Results and Impact: Our model strikes a balance between strong predictive performance and interpretability, while the web deployment enables accessible and transparent allergenicity assessment for both researchers and non-experts via our publicly available interface: xallergen.streamlit.app.

Challenges and Future Work: Remaining challenges include handling long sequences, incorporating structural features effectively, improving low sensitivity, outputting biologically relevant information such as identification of protein patterns common among allergens, and expanding the interface to support real-time 3D visualization and broader input compatibility.

1 INTRODUCTION

Allergies affect a significant portion of the global population, with 1–10% suffering from food allergies worldwide in 2023 [1], and 29.5% of Swiss adults self-reporting any allergy in 2022 [2]. Allergic reactions occur when IgE antibodies bind to specific epitopes on foreign proteins (allergens), triggering immune responses that range from mild discomfort to life-threatening anaphylaxis [3]. Experimentally, allergens are identified using enzyme linked immunosorbent assays (ELISAs) or liquid chromatography mass spectrometry (LCMS) methods [4]. While accurate, these techniques are time-consuming, costly, not always accessible and unsuitable for high-throughput screening—which is especially relevant in food safety and biotechnology.

To address this, various computational tools have been developed, such as AllerCatPro 2.0 [5], AllergyPred [6], AllergenAI [7], and AlgPred 2.0 [8]. Although they performed well on their respective prediction datasets, most lack an interactive user interface with interpretable prediction results or visualization. To combat this shortcoming, we developed a user-friendly and explainable allergenicity prediction tool based on protein sequences. We fine-tuned an ESM-2 protein language model with a feedforward neural network classification head. For model interpretability, we employed integrated gradients and an attention map. Our approach balances accuracy with interpretability, making it suitable for practical use and further analysis.

2 METHODS

2.1 Data Collection and Preprocessing

2.1.1 Protein sequence retrieval

The dataset used to train our models was sourced from AlgPred 2.0. The training dataset, curated by the developers of AlgPred 2.0, contains a balanced set of 8,060 allergenic protein sequences (positive samples) and 8,060 non-allergenic protein sequences (negative samples), totaling 16,120 sequences. The maximum and minimum sequence lengths in the training set are 3,326 and 50 amino acids, respectively. Among them, 14,986 sequences are no longer than 1,022 amino acids, making

them suitable for input into the ESM-2 model.

The test dataset also originates from AlgPred 2.0 and consists of 2,015 allergenic and 2,015 non-allergenic protein sequences, totaling 4,030 sequences. The sequence lengths range from 50 to 3,408 amino acids. Of these, 3,700 sequences are no longer than 1,022 amino acids and thus compatible with ESM-2’s input limit.

Detailed information about the dataset curation process, sources, and validation procedures is provided in [Appendix 6.1](#).

2.1.2 3D Protein Structure and 3D Structural Features Generation

To obtain 3D structural information for full-length protein sequences from the AlgPred 2.0 dataset [9], we employed ESMFold [10], a fast and accurate protein structure prediction model based on the ESM family. A total of 15,328 PDB files were generated for training and 3,947 for testing. ESMFold was run on an NVIDIA A100 GPU using Google Colab Pro+.

In addition, the full protein sequences were processed with DSSP [11] to extract structural features such as secondary structure elements, solvent accessibility, torsion angles, and hydrogen bonding patterns. The implementation of these 3D data is discussed in [3.3.2](#).

Although we also considered applying graph neural networks (GNNs) to exploit the 3D spatial topology, we ultimately ruled this out due to the high computational cost of GNN training on large datasets. For now, the predicted 3D structures are being used for downstream interpretability tasks, such as computing 3D attribution maps via integrated gradients [12], which are visualized through our user interface.

2.2 Usage of ESM-2 for Sequence Embedding

To represent protein sequences as high-dimensional embeddings for downstream classification, we utilized ESM-2, a state-of-the-art protein language model trained on a large and diverse set of sequences. ESM-2 captures structural and functional properties of protein sequences purely from their amino acid composition without requiring explicit alignment or structural input.

Due to an architectural limitation of ESM-2, input sequences are restricted to a maximum of 1,022 amino acids [13]. Consequently, we curated the AlgPred 2.0 dataset by removing all sequences exceeding this limit. This step introduced a slight class imbalance, which we addressed by further trimming excess samples from the majority class to ensure a balanced training and test dataset. After curation, the final dataset contained 14,986 protein sequences for training and 3,700 for testing.

Furthermore, to better adapt ESM-2 embeddings to the allergenicity prediction task, we also fine-tuned the ESM-2 model on the curated AlgPred 2.0 dataset, consisting exclusively of sequences with $\leq 1,022$ amino

acids. The embedding from the fine-tuned model was used as input to a neural network for classification. This fine-tuning improved model alignment with task-specific data and allows for better interpretability of the model.

On our website, we enforce an input constraint such that only protein sequences of length $\leq 1,022$ amino acids are allowed, maintaining compatibility with the ESM-2 model.

2.3 Model Training

To predict allergenicity, we extracted fixed-length embeddings from ESM-2 using both 320-dimensional `esm2_t6_8M_UR50D` and 1280-dimensional `esm2_t33_650M_UR50D` variants. These embeddings were used to train a range of models, including XGBoost, Ridge Regression, Feed-Forward Neural Networks (FFNN), Multilayer Perceptrons (MLP). We have also done a full fine-tuning of the smaller ESM-2 model `esm2_t6_8M_UR50D`. All models’ hyperparameters were selected using 5-fold cross-validation, either with Optuna, Keras Tuner or GridSearchCV, and evaluated using accuracy, sensitivity, specificity, precision, ROC-AUC, F1 score, and Matthews correlation coefficient (MCC).

2.3.1 Fine-tuning of ESM-2 with additional classification layer

We fine-tuned the foundation model `esm2_t6_8M_UR50D` with additional classification layers to predict protein allergenicity.

Hyperparameters, including hidden dimensions, number of layers, dropout rate, learning rate, batch size, and number of epochs, were optimized using Optuna with 5-fold cross-validation. The objective was to minimize validation loss. Early stopping was used to prevent overfitting and reduce unnecessary computation. After hyperparameter selection, the model was retrained on the full training dataset and evaluated on the holdout test dataset. Additionally, we conducted a Y-scrambling test as a sanity check to ensure the model’s predictive performance was not due to chance.

To accommodate limited computational resources, we used mixed precision training, which reduced memory usage and improved training speed. Despite using mixed precision training, the maximum batch size our computational resources (RAM of GPU) could use was eight. Therefore, larger batch sizes were not explored during hyperparameter tuning.

2.4 Model Evaluation

We evaluated our model’s performance using the independent test dataset provided by AlgPred 2.0 and further curated by us. To compare our models among each other, we reported several commonly used classification metrics (see [Appendix 6.2](#) for formulas).

Sensitivity (recall) was deemed the most important metric in allergenicity classification, because high sensitivity ensures that most true allergens are detected, which is crucial for safety in both food and pharmaceutical products. Other metrics that were regarded as favorable included high specificity, which would reduce product reformulation and cost by classifying fewer non-allergenic proteins incorrectly as allergenic. Moreover, high precision ensured that most proteins predicted as allergens were truly allergenic. Furthermore, F1 score balances precision and sensitivity, when accuracy is misleading in imbalanced datasets by penalizing one-sided performance. MCC considers all confusion matrix entries (true positives, true negatives, false positives, false negatives), making it suitable for safety-critical applications like allergenicity prediction, especially in imbalanced datasets, because it is not influenced by the proportion of positive and negative cases. In addition, we calculated the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which assesses the model’s ability to distinguish between positive and negative classes, reflecting model robustness without being tied to a decision threshold. In contrast, other metrics used are tied to the fixed threshold of 0.5, above which a prediction is classified as allergenic and below which it is classified as non-allergenic.

2.5 Explainable AI

Interpreting deep learning models is critical, especially in biomedical applications where trust and transparency are essential. To provide interpretability for our ESM-2-based allergenicity classifier, we applied two complementary techniques rooted in explainable AI (XAI): integrated gradients and attention analysis.

2.5.1 Integrated Gradients

We initially applied SHAP analysis (see Figure 3) to interpret the predictions of our XGBoost model. However, the results did not yield meaningful insights into how specific amino acids contribute to the allergenicity prediction result. SHAP analysis quantified the importance of each feature in the embeddings generated by ESM-2, but these high-dimensional embeddings are abstract representations learned by the model and do not directly correspond to individual amino acids.

Consequently, we changed our explainable AI strategy to Integrated Gradients (IG), as proposed by Sundararajan et al. [12]. IG was implemented in our fine-tuned ESM-2 model to enable more biologically meaningful interpretation by directly attributing predictions to individual amino acid residues in the input sequence.

Integrated Gradients is a feature attribution method grounded in two axiomatic principles: *Sensitivity*, which ensures relevant features are assigned non-zero importance, and *Implementation Invariance*, which guaran-

tees consistent attributions across functionally equivalent models, making it suitable for interpreting complex models like transformers.

The method computes the integral of gradients of the model’s output with respect to the input, along a linear path from a baseline input to the actual input:

$$\text{IG}_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha \quad (1)$$

where x is the input, x' is the baseline, and F is the model function.

In our implementation, we selected an all-zero embedding as the baseline and computed the attribution score for each amino acid in the input sequence. These scores reveal the extent to which individual residues contribute to the model’s prediction of allergenicity. IG was particularly useful for highlighting residues that can potentially be linked to epitopes or other biological features, thus enhancing the interpretability of the model’s decision-making process.

2.5.2 ESM-2 Attention Map

In addition to IG, we examined the attention mechanism intrinsic to the transformer architecture of ESM-2. Attention weights quantify the dependencies between tokens in a sequence and are used by the model to selectively aggregate information across positions. We extracted the attention matrices from the final transformer layer and visualized them as heatmaps. These attention maps help elucidate which residues the ESM-2 model focuses on when encoding a given amino acid sequence, offering insight into the internal representation learned by the model, though attention weights do not constitute a formal explanation that are biologically significant. Notably, these attention patterns come from ESM-2 and are independent of the classifier’s predictions.

2.5.3 3D Attribution Mapping

To gain spatial insight into the model’s decision-making process, we computed residue-level attribution scores utilizing the integrated gradients method. These raw attribution scores, which can be both positive and negative, reflect the degree and direction of each amino acid’s contribution to the final allergenicity prediction.

For visualization, we first normalized the values of the attribution scores to the range $[-1, 1]$ and injected them into the B-factor field, a metadata field in the PDB format, of the corresponding PDB files. This enabled attribution scores to be mapped directly onto the 3D protein structures and visualized with molecular viewers such as py3Dmol. In py3Dmol, residues with high positive contributions (supporting allergenicity) were colored in blue, while residues with strong negative contributions

(suppressing allergenicity) were colored in red based on the magnitude of their attributions.

This 3D attribution mapping provides an intuitive and spatially resolved visualization of which regions of a protein sequence are most critical for the model’s prediction, offering both interpretability and potential biological insights.

This feature is currently limited to proteins for which 3D structures have been precomputed using ESMFold. Real-time 3D structure generation for arbitrary user-input sequences on the web interface is currently not supported due to the high computational cost. Future work may investigate lighter-weight structure prediction pipelines to enable broader access to real-time structural interpretability.

2.6 Dummy Classifier as Baseline

To evaluate the relative performance of our trained models, we implemented a stratified dummy classifier that generated random predictions while preserving the original class distribution. This baseline allowed us to quantify the predictive signal present in the embedded sequences and assess whether our model was learning meaningful patterns beyond random chance.

2.7 Web Interface

We developed an interactive web application using Streamlit to make our model accessible and interpretable. The application is publicly deployed at <https://xallergen.streamlit.app>, providing free access to researchers and practitioners worldwide. Users can input one or more protein sequences and receive real-time predictions of allergenicity, presented in a summary table showing both binary labels and probabilities. Currently the website is on a free plan and might take eight minutes to wake up after falling asleep.

To support model interpretability, users can select an individual sequence and view a bar plot of integrated gradients, highlighting which residues contributed most to the prediction. Positive (blue) and negative (red) attributions help interpret model behavior at the amino acid level. Beneath the plot, a ranked table lists the most influential residues based on attribution magnitude, offering a clearer overview of key sequence positions. If a PDB file is available, the corresponding 3D structure is visualized using py3Dmol, with residue-level attributions mapped onto the protein’s structure. Additionally, the app supports optional visualization of transformer attention maps for further interpretability. Screenshots of the interface are shown in [Appendix 6.7.3](#).

2.8 Workflow Overview

The overall project workflow, from raw data to model training, performance evaluation and explainability, is illustrated in Figure 1.

3 RESULTS AND DISCUSSION

3.1 Model Comparisons

The different models were evaluated on the test set and compared in terms of accuracy, sensitivity, specificity, ROC-AUC, F1 score, and MCC (Figure 2).

All models showed promising results and outperformed the dummy model, with especially the Ridge Classifier with 320-dimensional ESM-2 embeddings, the XGBoost model with 1280-dimensional ESM-2 embeddings, and the XGboost + MLP ensemble model with 1280-dimensional ESM-2 embeddings standing out. However, the fine-tuned ESM-2 model was chosen as the final predictor for its interpretability. Although its sensitivity value was lower than that of the best models, its remaining performance metrics were satisfactory. In particular, the model achieved a high specificity score, indicating that it made very few false positive predictions.

3.2 Model Benchmarking

We selected two models for a detailed discussion: the fine-tuned ESM-2 model, chosen for its model interpretability, and the MLP + XGBoost ensemble model with 1280-dimensional embeddings from ESM-2, selected for achieving the highest overall performance metrics on the test set. Performance results are shown in Table 1.

Comparing the two models internally, they perform similarly across all key metrics, with the ensemble model using ESM-2 1280-dimensional embeddings consistently scoring higher. Both models exhibit exceptionally high specificity, 97.9% for the fine-tuned ESM-2 and 98.8% for the ensemble model, indicating very few false positives. This low false positive rate suggest that when the model predicts a protein as allergenic, the prediction is reliable and unlikely to be an incorrectly classified non-allergen. However, their sensitivity is noticeably lower, at 70.1% and 77.8%, respectively, suggesting that a proportion of true allergens are missed, particularly by the fine-tuned model. The ensemble model also achieved higher overall accuracy (88.3% vs. 84.0%), a stronger ROC-AUC (98.1% vs. 91.9%), and a better Matthews correlation coefficient (78.3% vs. 70.8%), indicating the ensemble model’s prediction performance is better in all four confusion matrix categories.

Although the model’s sensitivity can be improved, it usually comes with the cost of lowering specificity, as can be seen in other existing models, like AllerCatPro 2.0 [5]. A key advantage of our fine-tuned ESM-2 model is its built-in interpretability, which sets it apart from existing allergen prediction methods that largely function as black-box models.

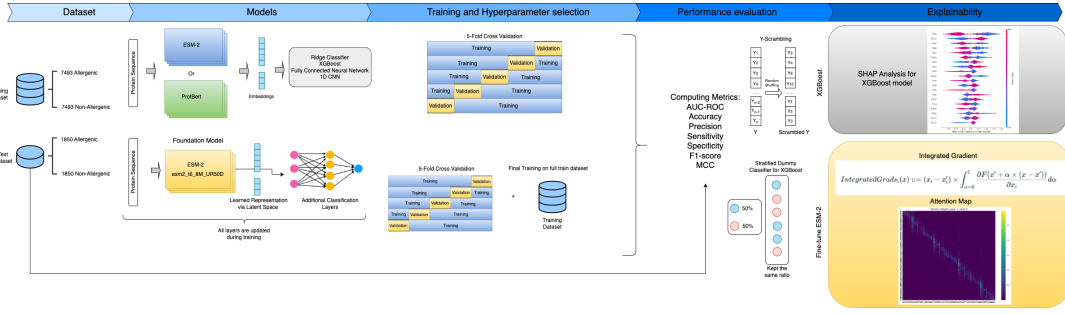


Figure 1: Workflow overview: Data collection, model architecture, model training and hyperparameter selection, performance evaluation, and explainable AI

Table 1: Comparison of two selected allergenicity prediction models across key evaluation metrics.

Model	Accuracy	Sensitivity	Specificity	ROC-AUC	MCC
Fine-tuned ESM-2	84.0%	70.1%	97.9%	91.9%	70.8%
XGBoost + MLP Ensemble ESM-2 1280-dim	88.3%	77.8%	98.8%	98.1%	78.3%

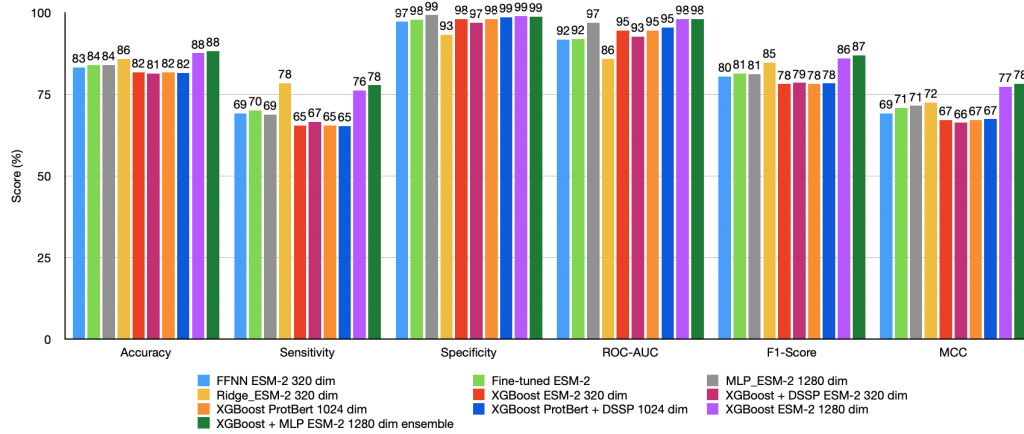


Figure 2: Comparison of different metrics between tested models. Accuracy, sensitivity, specificity, F1 score, ROC-AUC, F1 score and MCC are depicted from left to right.

3.3 Limitations

3.3.1 ESM-2 based models

Although the ensemble model achieved the highest overall performance across all metrics, it was not chosen as the final predictor. This is because it operates as a black-box model without built-in interpretability. In contrast, the fine-tuned ESM-2 model incorporates interpretability through integrated gradients and attention maps, enabling insights into residue-level contributions. Never-

theless, its lower sensitivity (70.1%) means it fails to identify nearly one-third of true allergens, indicating that potential for improvement remains. Finally, both the fine-tuned ESM-2 and XGBoost models are limited by the ESM-2 architecture's maximum input length of 1,022 amino acids, restricting predictions of longer protein sequences.

3.3.2 Integration of 3D information

We tried to concatenate the 3D features calculated by DSSP with the precomputed high-dimensional ESM-2 embeddings as additional input features. However, this fusion led to a decrease in model performance. Upon investigation, we identified a dimensionality mismatch between the low-dimensional DSSP features and the high-dimensional ESM-2 embeddings as the root cause of this performance degradation.

To address this, we developed CAMCNN, which learns from one-hot encoded protein sequence using 1D-CNN and from 3D structural features using a fully connected neural network. The latent representations from both branches are then concatenated for binary classification with dynamic thresholds (details see in 6.6), aiming to preserve interpretability while improving classification accuracy. However, this approach resulted in a low accuracy, sensitivity, MCC, and F1 score despite high precision and AUC. We suspect that the initial separation and late stage fusion of sequence and structural data could lead to imbalanced feature contributions and overfitting. Future work should explore earlier fusion, ideally before loading into the CNN. Moreover, excessive overpadding may dilute the model's performance.

4 CONCLUSION AND OUTLOOK

In this project, we successfully developed XAllergen, an explainable AI framework for predicting protein allergenicity that addresses a critical gap in computational allergenicity assessment. Our approach combines the power of the ESM-2 protein language model with interpretable machine learning techniques to deliver both accurate predictions and meaningful biological insights.

Our fine-tuned ESM-2 model achieved 84.0% accuracy with exceptionally high specificity (97.9%), indicating that when an allergen prediction is made it is unlikely to be an incorrectly classified non-allergen. While the sensitivity (70.1%) indicates room for improvement in detecting all true allergens, the model's interpretability through integrated gradients and attention mechanisms provides unprecedented transparency in allergenicity prediction. This explainable approach represents a significant advancement over existing black-box models in the field, enabling researchers to understand which amino acid residues drive allergenicity predictions.

The interactive Streamlit application, now publicly accessible at <https://xallergen.streamlit.app>, provided an accessible and user-friendly interface that successfully democratizes access to allergenicity prediction without requiring specialized computational expertise. The integration of sequence analysis with 3D structural visualization offers users comprehensive insights into protein allergenicity from multiple perspectives. Interpretability enhances trust in computational allergenicity predictions

by clarifying which amino acid residues drive model decisions — critical when experimental validation is impractical. Such transparency aids protein engineering by guiding sequence modifications, supports regulatory assessments with justifiable risk explanations, and facilitates model improvement through bias and artifact detection.

While direct benchmarking against established tools like AlgPred 2.0, AllerCatPro, and AllergenAI is not feasible due to our curated dataset, our model's key differentiator lies in the built-in interpretability that enables users to understand and trust the predictions. The trade-off between sensitivity and specificity in our model reflects a deliberate design choice to minimize false positive predictions while maintaining reasonable detection capability, with the added benefit of transparency in the decision-making process.

4.1 Future Work and Improvements

Several promising directions emerge from this work that could significantly enhance the system's capabilities and broader applicability:

Overcoming Sequence Length Limitations: A primary limitation of our current approach is the ESM-2 architecture's restriction to sequences of ≤ 1022 amino acids, which excludes many biologically relevant proteins. Recent advances in protein language model architectures [13] offer promising solutions for extending sequence length capabilities.

Systematic Allergenic Motif Detection : A particularly promising direction involves applying our interpretable pipeline to analyze all allergenic proteins within specific protein families to detect characteristic allergenic motifs unique to each family. By systematically examining the integrated gradient attributions and attention patterns across multiple allergens from the same protein family, we could identify conserved residues and sequence regions that consistently contribute to allergenicity predictions. This family-specific analysis approach could reveal distinct allergenicity signatures that enhance both prediction accuracy and potentially biological interpretability. Incorporating additional biological information such as structural motifs, hydrophobicity profiles, post-translational modifications, or organism taxonomy could further improve prediction accuracy.

Addressing Low Sensitivity : In the future, we should investigate techniques to improve sensitivity and investigate sensitivity-specificity trade-offs to obtain a more balanced performance of the model.

Improved 3D Structure Integration and Real-Time Visualization : Our attempts to integrate 3D structural features with sequence embeddings revealed challenges in effectively combining low-dimensional struc-

tural descriptors with high-dimensional ESM-2 embeddings. Future work should explore more sophisticated fusion strategies, such as early-stage integration before model training, attention-based feature weighting, or developing specialized architectures designed for multi-modal protein data. Another particularly exciting development would be enabling real-time 3D attention mapping of any user-input protein sequence using ESMFold or lighter computational variants directly on our website. This would allow users to visualize attention patterns and attribution scores on real-time generated 3D structures for any protein sequence beyond the current precomputed structure library.

Expanding Interpretability Methods : While integrated gradients and attention mechanisms provide valuable insights, incorporating additional explainability techniques such as counterfactual explanations could offer complementary perspectives on model decision-making. Developing allergen-specific visualization tools that highlight known epitope regions or conserved motifs would enhance biological interpretability.

Broader Deployment and Validation : Our web interface is now publicly accessible at <https://xallergen.streamlit.app>, making it broadly available to researchers, practitioners, and regulatory agencies worldwide. Future enhancements will include additional models with varying sensitivity-specificity trade-offs for users to choose depending on their specific needs. Conducting prospective validation studies with experimental allergenicity testing would strengthen confidence in the model’s real-world applicability. Expanding the interface to support batch processing, API integration, and additional file formats would enhance its utility for high-throughput applications in food safety and pharmaceutical development.

Cross-Platform Integration : Developing plugins or integrations with popular protein analysis platforms and databases would facilitate adoption within existing bioinformatics workflows.

Ultimately, our framework illustrates the feasibility of integrating protein language models with interpretable machine learning for predictive tasks in computational biology. As the field moves towards more transparent and explainable AI systems, tools like XAllergen will play increasingly important roles in bridging the gap between computational predictions and biological understanding, ultimately contributing to safer food production, pharmaceutical development, and biotechnology applications.

5 CONTRIBUTION CONFIRMATION

We, the undersigned, confirm that we have all contributed significantly to the planning, implementation,

and documentation of this project. We have read and approved the final version of the report, and we agree with its content and submission.

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6 APPENDIX

6.1 AlgPred 2.0 Dataset: Detailed Description and Curation Process

The AlgPred 2.0 dataset comprises experimentally validated allergens and carefully selected non-allergens from multiple major repositories, including COMPARE [14], Allergen Online [15], AllerTOP [16], AlgPred [8], and Swiss-Prot [17].

The allergens are known to cause IgE-mediated immune responses in humans and include proteins derived from a wide variety of organisms such as plants (e.g., pollen, nuts, fruits), animals (e.g., milk, eggs, fish, shellfish, insects, mites), fungi (e.g., molds, yeasts), bacteria, and other sources such as latex. Non-allergens were extracted from Swiss-Prot using a strict query (NOT allergen NOT cancer NOT allergenic AND reviewed: yes) to ensure no overlap with allergens, and only reviewed, manually annotated proteins were retained. The dataset was carefully curated to include 10,075 allergens and 10,075 non-allergens. Proteins containing fewer than 50 amino acids or non-standard characters were excluded, and redundancy was minimized using CD-HIT clustering at a 40% sequence identity threshold applied independently to allergen and non-allergen sets [18]. The resulting clusters were randomly partitioned into training and testing sets in an 80:20 ratio, ensuring that no protein in different subsets shared high sequence similarity, thus promoting generalizability.

In addition to sequence-level curation, IgE epitope mapping was performed using data from IEDB, AllerBase, and IgPred, yielding 10,451 IgE epitopes and 307,866 non-IgE epitopes after removing redundancy and applying length filters. Two techniques were used for epitope identification: BLAST-based searching and motif-emerging classes identification (MERCİ). The MERCİ software was specifically applied to identify motifs present in experimentally validated IgE epitopes, and a protein was labeled as an allergen if it contained a motif identified by MERCİ or matched a known IgE epitope. This comprehensive and rigorously filtered dataset underpins the robust allergenicity prediction capabilities of AlgPred 2.0.

6.1.1 Initial Experiments with IEDB Dataset

Prior to adopting the AlgPred 2.0 dataset, we initially experimented with training our model using protein epitope sequences downloaded from the IEDB database [19]. The IEDB-derived dataset included 5,293 positive samples (epitopes) and 7,021 negative samples. Epitope lengths ranged from 3 to 61 amino acids for positives, and from 4 to 56 amino acids for negatives.

However, the model trained on this dataset exhibited relatively low predictive performance. Upon investigation, we found that the data lacked proper curation and

suffered from class imbalance. Also, the epitope sequences are often very short and lack spatial context. These limitations likely led to overfitting and poor generalization. As a result, we transitioned to the better-curated and CD-HIT-filtered AlgPred 2.0 dataset to ensure improved robustness and performance.

6.2 Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{F1 score} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (6)$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (7)$$

6.3 Best Performing Model: MLP + XGBoost Ensemble on ESM-2 Embeddings

Although not selected as the final deployed model, our best-performing configuration during development was a soft-voting ensemble combining a Multi-Layer Perceptron (MLP) and an XGBoost classifier. Both models were trained on 1280-dimensional protein embeddings obtained from the ESM-2 (T33 650M) model, mean-pooled across residues.

6.3.1 Model Configuration

- *ESM-2 embeddings*: Each protein sequence from AlgPred 2.0 was converted into a 1280-dimensional feature vector using mean pooling of final-layer token embeddings.
- *CNN*: A 1D convolutional neural network was used for embedding-based classification. Hyperparameters were optimized using Keras Tuner.
- *MLP*: Two hidden layers (192 and 64 units), ReLU activation, dropout 0.2, trained with Adam optimizer (learning rate = 0.0004).
- *XGBoost*: Tuned via randomized search across learning rate, max depth, and regularization parameters. Feature importance was used for interpretability.
- *Ensemble*: A soft-voting ensemble of MLP and XGBoost models, averaging predicted probabilities from both classifiers.

6.3.2 Performance Evaluation

- *5-fold Cross-validation:* ROC-AUC = 0.946 ± 0.006 , Accuracy = 0.885 ± 0.005
- *Independent Test Set:* ROC-AUC = 0.951, Accuracy = 0.891
- *Y-Scrambling Control:* Randomly shuffled labels resulted in a ROC-AUC of ca. 0.51, confirming that the model learned meaningful patterns.

6.4 Supplementary material for XGBoost model using a 1280 dimensional ESM-2 embedding

6.4.1 Hyperparameter selection with cross validation (CV)

GridSearchCV library from sklearn was used to select the best hyperparameters using 5-fold CV. The hyperparameters tuned are max depth, learning rate, number of estimators, subsample size, and colsample_bytree (fraction of features (columns) to be randomly sampled for each tree during training).

Table 2: Cross-Validation Performance of Best Model

Metric	Value
Mean CV ROC-AUC	0.9927 ± 0.0070

The best hyperparameters are:

```

1 # Best model
2 best_model = xgb.XGBClassifier(
3     use_label_encoder=False,
4     eval_metric="logloss",
5     random_state=42,
6     colsample_bytree=0.8,
7     learning_rate=0.1,
8     max_depth=5,
9     n_estimators=200,
10    subsample=1.0
11 )

```

6.4.2 Y-scrambling

Y-Scrambled ROC-AUC: 0.4903 ± 0.0128

6.4.3 SHAP

SHAP analysis result of the model XGBoost using 1280 dimension embedding from ESM-2 as input.

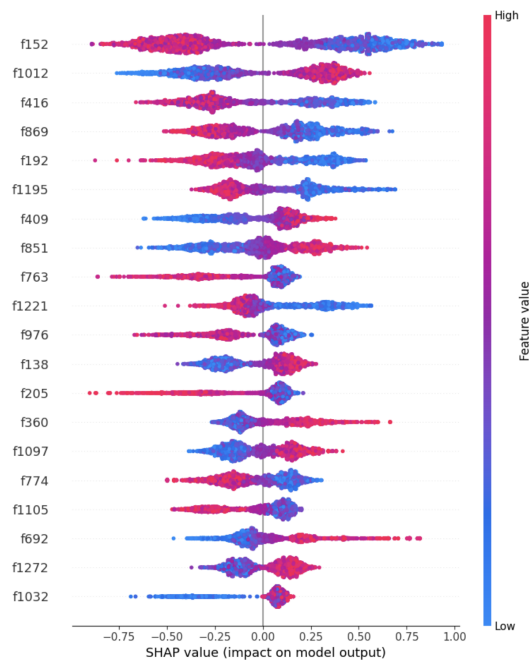


Figure 3: SHAP analysis of the model XGBoost with a 1280 dimensional embedding from ESM-2

6.5 Ensemble Model: MLP + XGBoost Soft Voting

To leverage the complementary strengths of different model architectures, we developed an ensemble approach combining our best-performing individual models through soft voting. The ensemble integrates predictions from an MLP and an XGBoost classifier, both trained on 1280-dimensional ESM-2 embeddings.

6.5.1 Ensemble Methodology

The ensemble employs a soft-voting strategy that averages the predicted probabilities from both component models:

$$P_{\text{ensemble}}(y = 1|x) = \frac{P_{\text{MLP}}(y = 1|x) + P_{\text{XGBoost}}(y = 1|x)}{2} \quad (8)$$

where $P_{\text{ensemble}}(y = 1|x)$ represents the ensemble probability of allergenicity for input sequence x . Final binary predictions are made using a threshold of 0.5 on the averaged probabilities.

6.5.2 Component Models

MLP: A deep neural network with three hidden layers (256, 128, 64 units) featuring ReLU activations,

dropout regularization (0.3, 0.3, 0.2), and trained using the Adam optimizer with a learning rate of 0.0005. The model achieved strong performance with high specificity and balanced sensitivity.

XGBoost Classifier: A gradient boosting model optimized through randomized search with the following best hyperparameters: 400 estimators, maximum depth of 6, learning rate of 0.05, subsample ratio of 0.9, and colsample_bytree of 1.0. The model was integrated into a scikit-learn pipeline with standardized input features.

6.5.3 Performance Evaluation

The ensemble model was evaluated alongside individual component models using comprehensive metrics on the independent test set. Figure 4 presents the ROC curves for all models, demonstrating the discriminative performance across different operating points.

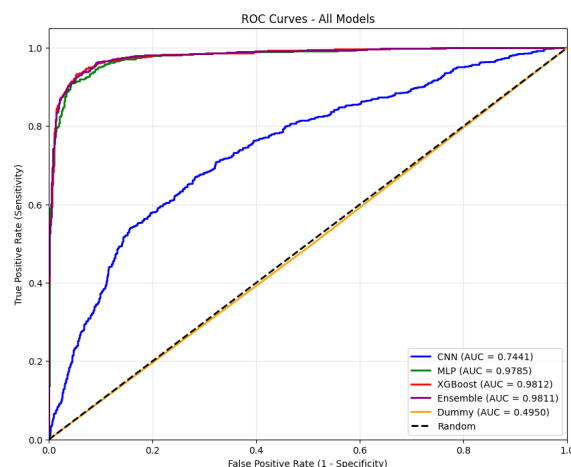


Figure 4: ROC curves comparing all models on the test set. The ensemble model (purple) achieves strong discriminative performance (AUC = 0.9811) that closely matches the best individual models while providing improved stability through model averaging.

Table 3 and Figure 5 present the comprehensive performance comparison between individual models and the ensemble on the independent test set.

6.5.4 Analysis and Discussion

The ensemble model demonstrates balanced performance that generally falls between the individual component models, effectively capturing their complementary strengths. Key observations include:

- **Robust Performance:** The ensemble achieves 88.3% accuracy with 98.8% specificity, maintain-

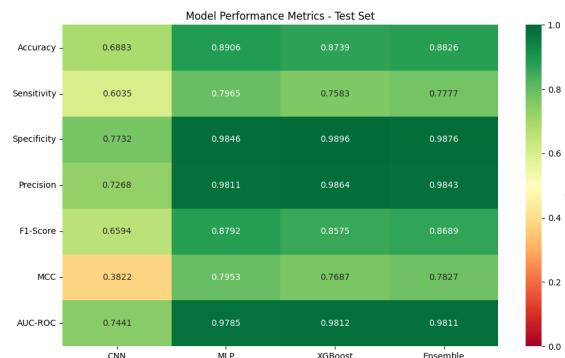


Figure 5: Heatmap visualization of model performance metrics on the test set. The color intensity represents metric scores from 0 (red) to 1 (dark green). The ensemble consistently achieves balanced performance across all metrics, with particularly strong specificity and precision scores.

ing the high reliability characteristic of both component models for confirming non-allergenic proteins.

- **Sensitivity-Specificity Balance:** With 77.8% sensitivity, the ensemble provides a middle ground between the higher sensitivity of MLP (79.7%) and the lower but more conservative sensitivity of XGBoost (75.8%).
- **Maintained Precision:** The ensemble preserves excellent precision (98.4%), ensuring that positive allergenicity predictions remain highly reliable.
- **Stability:** The MCC of 78.3% and AUC-ROC of 98.1% indicate robust discriminative performance, with the ensemble providing more stable predictions through averaging.

While the ensemble does not substantially outperform the best individual model (MLP), it offers improved prediction stability and reduced variance via model averaging. This approach is particularly valuable in safety-critical applications where consistent and reliable predictions are paramount. The ensemble methodology validates that both neural network and tree-based approaches have learned complementary aspects of the allergenicity prediction task from ESM-2 embeddings.

6.6 Allergen Prediction using CAMCNN: Methodology and Results

A Convolutional Neural Network model with Class Activation Map (CAMCNN) was developed and evaluated

Table 3: Performance comparison of individual models and ensemble on the test set.

Model	Accuracy	Sensitivity	Specificity	Precision	F1 score	MCC	AUC-ROC
MLP	89.1%	79.7%	98.5%	98.1%	87.9%	79.5%	97.9%
XGBoost	87.4%	75.8%	99.0%	98.6%	85.8%	76.9%	98.1%
Ensemble	88.3%	77.8%	98.8%	98.4%	86.9%	78.3%	98.1%

for the task of allergen prediction from protein sequences and structural features. This subsection details its architecture, training protocol, and performance.

6.6.1 Model Architecture (CAMCNN)

The CAMCNN model employs a hybrid architecture:

- **Sequence Processing Branch:** One-hot encoded protein sequences (alphabet size 20) are processed by a 1D convolutional layer (128 filters, kernel size 16), followed by Batch Normalization, a ReLU activation, and 1D Max Pooling (kernel size 2, stride 2).
- **Structural Feature Branch:** Averaged structural features (10 per residue) are input to a fully connected layer (128 units), followed by Batch Normalization and ReLU activation.
- **Feature Fusion and Classification:** Outputs from both branches are concatenated and passed through two dense layers. The first has 256 units (ReLU), and the final output layer uses a single unit with a sigmoid activation for binary classification.

The model is designed to support Class Activation Mapping (CAM).

6.6.2 Training and Evaluation Protocol

Dataset and Preprocessing: Preprocessed one-hot encoded sequences and structural features (.npz files, 32-bit float) were used, split into training and hold-out test sets.

Nested Cross-Validation: Performance was estimated using a 5-fold outer (stratified) CV on the training set. Within each outer fold, a 3-fold inner (stratified) CV determined the optimal training epochs (maximum 100, patience 10) by maximizing validation AUC. The average optimal number of epochs was 23.2 ± 3.5 .

Training Parameters: The Adam optimizer (learning rate 1×10^{-3}) and BCEWithLogitsLoss (with class weights for imbalance) were used. Batch size was 32. A per-fold optimal threshold was selected from the validation precision-recall curve to maximize F1 score. A random seed (42) ensured reproducibility.

Final Model: The CAMCNN was retrained on the full training set for 23 epochs (average from CV) and evaluated on the test set using thresholds from two strategies: (1) average validation threshold (cv_threshold), and (2) optimal F1 score threshold (optimal_threshold). A DummyClassifier (stratified) served as a baseline during CV.

6.6.3 Performance Metrics

Evaluation metrics included ROC-AUC, accuracy, F1 score, MCC, precision, and recall.

Nested Cross-Validation Performance: Table 4 shows the mean and standard deviation of metrics from the 5-fold nested CV for CAMCNN and the Dummy classifier.

Hold-out Test Set Performance: The final CAMCNN model’s performance on the independent test set using two thresholding strategies is shown in Table 5.

6.7 Supplementary Information for fine-tuning of ESM-2 model

6.7.1 Cross-Validation Performance of Best Trial

The average cross-validation loss of the best Optuna trial was 0.1154 ± 0.0095 .

Best Hyperparameters:

```

1 best_params = {
2     'hidden_dim': 128,
3     'num_layers': 2,
4     'dropout': 0.10582998499416787,
5     'lr': 8.293674908605468e-06,
6     'batch_size': 4,
7     'max_epochs': 7 # The early stopping has an
8     ↪ average of 7 epochs
9 }
```

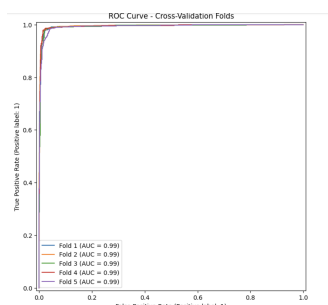
Table 4: Nested CV Performance (Mean $\pm$ SD).

Model	AUC	Accuracy	F1 score	MCC	Precision	Recall
CAMCNN	0.9959(10)	0.9853(10)	0.9859(10)	0.9707(20)	0.9938(17)	0.9780(14)
Dummy	0.5000	0.5232(1)	0.0000	0.0000	0.0000	0.0000

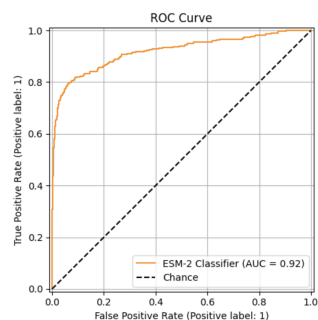
Table 5: Final CAMCNN Performance on Hold-out Test Set.

Threshold Strategy	AUC	Accuracy	F1 score	MCC	Precision	Recall	Threshold
CV Threshold	0.8806	0.6662	0.5216	0.4416	0.9690	0.3569	0.6392
Optimal Threshold	0.8806	0.7174	0.6316	0.5085	0.9419	0.4751	0.1000

6.7.2 ROC-AUC curve for 5-fold CV and hold-out test set



(a) 5-fold CV for hyperparameter tuning



(b) Hold-out test set performance

Figure 6: AUC-ROC curves for fine-tuned ESM-2 on (a) 5-fold CV during hyperparameter tuning and (b) the hold-out test set.

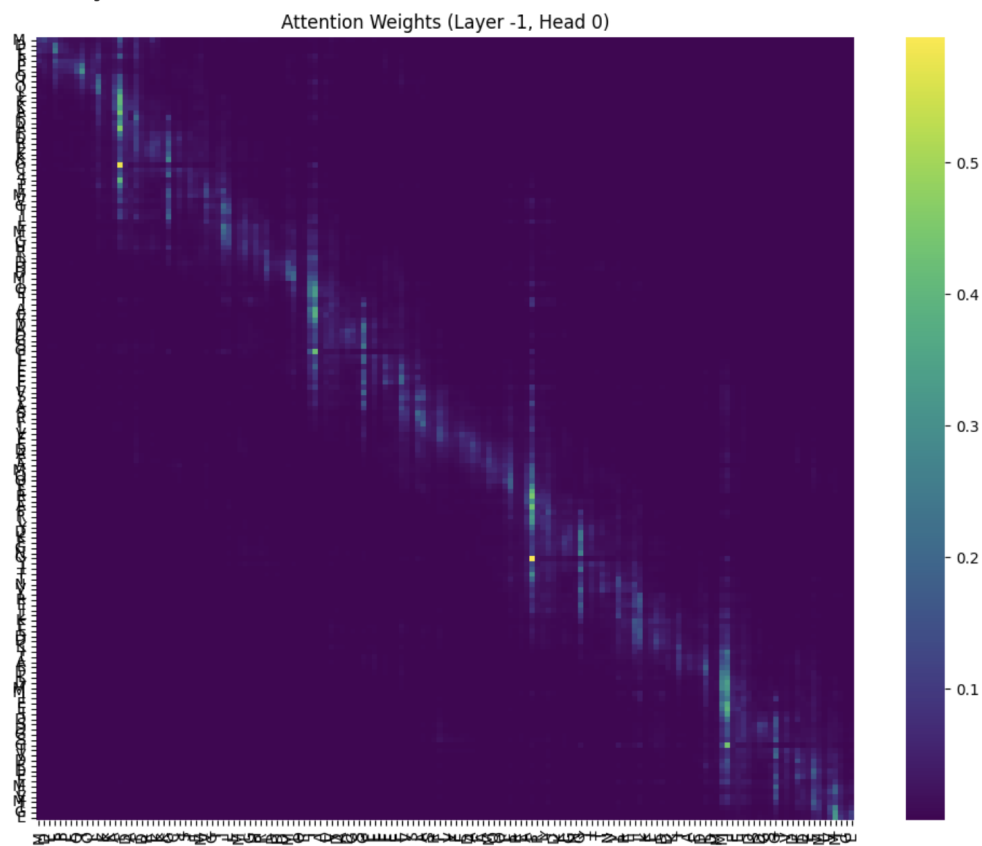
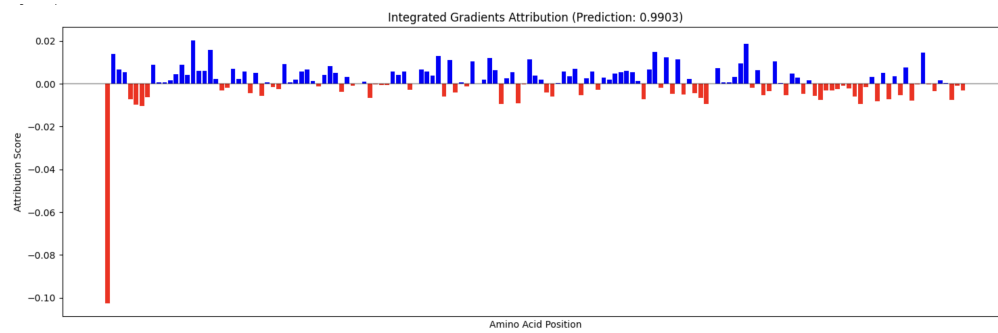
6.7.3 Interpretability

Figure 7 shows the analysis attribution score using the integrated gradient method for a protein sequence. Figure 8 shows the attention map from the attention

weight of the last layer of the ESM-2 model. Figure 9 shows a screenshot of the entire XAllergen web interface.

6.8 Additional Information

Additional information, documentation, and the user interface application are available on the project's [GitHub repository](#).



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The authors take full responsibility for the accuracy and originality of the work presented.